

Project Phase 2: Data Layer

TIM 175, Spring 2026

May 31, 2026

Business Case

1.1. Define the Problem

Decision: Erin Walsh the CPO has a strategy review in 30 days where she will describe if the Rollouts feature is providing enough return on investment so that the updated pricing in the Pro or Enterprise tier is increasing revenue in the company. To achieve this Erin needs 3 numbers to walk into this meeting with:

1. Count of orgs whose active subscriptions at month end is Pro or Enterprise from January 2026 or later.
2. Percent of orgs currently on Pro+ at the most recent month end, April 2026, and the share that have fired a rollout notification January 2026 onward.
3. Monthly MRR by plan tier from January 2026 onward.

Baseline: Today there is no analytics layer. Erin's analyst pulls the numbers by hand from the four source systems (billing, tickets, notifications, and identity) and assembles key numbers by hand. Each pass takes a few days and starts from scratch when answering each different question. So far the process has not produced a single number Erin can defend.

Cost of getting it wrong: A wrong number will lead to decisions being made in the wrong direction. Both underreporting and overreporting will lead to investment being put in the wrong way and non-optimal decisions will be made.

- **Underreporting:** would lead to the CPO rolling back investment or potentially cutting the feature completely.
- **Overreporting:** would lead to the CPO defending the value of the Rollouts feature when it does not provide real value to customers, which would lead to user churn.

Cost of not doing it: The current timing of two days for each pass will continue, which creates data that is error prone and is a static source that will need to be recreated every time there is a question related to the analytics of the organization. This current system produces numbers that the CPO would not be able to defend, leading to decisions being

made blindly and outcomes that could have been resolved if there was work done on the analytics layer to answer important business decisions.

1.2. Choose the Approach

Simplest approach that clears the bar: We have decided to implement an in-house data system that builds the star schema on top of the built-in source tables already inside the warehouse. We will introduce new variables that will store the necessary information to answer the questions the CPO has requested.

What we rejected:

- **Current system:** No analytical layer which does not scale or provide data that the CPO can defend.
- **Queries to the operational database:** Creating a system that completed individual queries will lead to a system of inaccuracy, as data will contain only current state with no historical accuracy.
- **Third party data processing:** We already have a solid base of data inside the warehouse and the ability for the firm to own the analytics layer will allow them to answer questions specifically how they need it.
- **System that overachieves:** A system that is overbuilt which would help answer future questions and provide stronger integration. This is rejected because the CPO needs reliable numbers in 30 days and a project such as this would push the timeline back. Waiting longer to have reliable information will lead to the business making decisions blindly.

1.3. Justify Value

Implementing a working analytical layer is a commodity, and it provides enough value to implement an in-house standardized data system. The implementation that is described will connect the business segments in a standardized dimension which can be extended to answer other similar business decisions. Considering this system will be built quickly and without significant modifications, the questions it answers would be standard to what comparable companies would be asking. The questions that are needed in this build are not complex business intelligence questions but instead standard to understand how your business is operating.

1.4. Success Criteria

1. The three required statistics are produced from the data model in a timely manner.
2. The numbers reconcile to the same amounts throughout the schema.
3. Erin treats the data as defensible amounts.
4. The analytics layer is continued to be used even after the strategy review.

5. Erin's analyst stops doing manual pulls for data.

1.5. Plan Adoption

This new plan will be implemented in the CPO's strategy review deck which is occurring in 30 days. With this new analytics layer implementation it cuts out the two-day manual query searching, leading to less work for Erin's analyst. Reliable questions can now be answered quickly. This is beneficial because if the strategy review requires follow-up questions, this meeting can be the opportunity to solve those issues. In the future this same analytics layer can be used to get up-to-date updates on the same questions through a Rollouts dashboard, allowing for less work when continuing to ask the same questions moving forward, such as a source of truth for board updates and the basis for the Pro+ renewal narrative at year end.

Star Schema Design

2.1. Schema Diagram

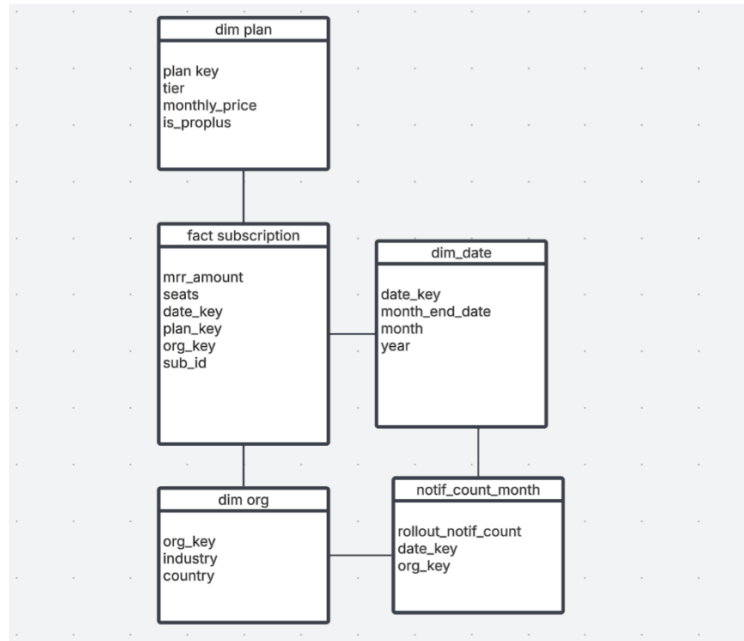


Figure 1: Star schema: `fact_subscription` and `notif_count_month` with `dim_plan`, `dim_org`, and `dim_date`.

Grain Level: Monthly. This gives one row per active subscription per month end, one row per org per month. Questions 1 and 3 explicitly ask for month-end snapshots, so having a grain at this level makes it something that can be naturally created as a snapshot without something that you have to derive through every specific read.

Rejected Grain: Event grain. This level was rejected because this grain would not allow you to directly create queries related to the facts at month end, leaving you at a similar point as if you did not alter the data.

Measures picked:

- **mrr_amount: semi-additive** ($\text{mrr_amount} = \text{monthly_price} * \text{seats}$) — Adds across subscriptions and tiers within a month but not across multiple months. The reason it is not summed across months is because that would introduce double counting.
- **seats: semi-additive** — Since seats tracks the number of seats in a current moment, they represent the same population over time, so summing across months would also introduce double counting for individuals.
- **rollout_notif_count: additive** — This amount is additive across all dimensions. The purpose of this is to calculate a summation of all notifications throughout the period.

Why: Each fact row represents only the measure which matches its respective grain. For example, MRR and seats are on the subscription snapshot which cannot be summed across time due to them being levels, whereas rollout notification count is on the rollout fact and can be summed across time.

Rejected measures:

- **Aggregate MRR by tier:** No point storing this variable because it can be computed as a query later.
- **Active flag:** Every row is already active at month end if it is present in the table. No need for an active flag.

Dimensions:

- **dim_plan:** Stores the plan tier and pricing information for each subscription. Attributes: `plan_key`, `tier`, `monthly_price`, `is_proplus`.
- **dim_date:** Default date dimension that handles tracking the month end for each period. Attributes: `date_key`, `month_end_date`, `month`, `year`.
- **dim_org:** Holds organizational attributes. Attributes: `org_key`, `industry`, `country`.

Rejected dimensions:

- **dim_user:** Rejected because it does not answer any of the necessary questions for this specific business case.
- **dim_billing_cycle:** Rejected because in our business case monthly and annual subscriptions are treated identically.

Analysis

Q1: Monthly Pro+ Subscribers

The question: Count of orgs whose active subscription at month-end is Pro or Enterprise, from January 2026 onward.

Grain: One row per active subscription per month-end.

Model:

- **fact_subscription:** monthly snapshot fact, reconstructed from `billing_subscription_changes`.
- **dim_plan:** one row per plan. `plan_key`, `tier`, `monthly_price`, `is_proplus`.
- **dim_org:** one row per org. `org_key`, `industry`, `country`.

How the reconstruction works:

- A subscription is active at month-end if it has a **created** event on or before month-end and no **cancelled** event on or before month-end.

- Plan at month-end is the `to_plan.id` of the most recent `created` or `plan_changed` event on or before month-end.
- Uses the change log, not current `billing.subscriptions`, so historical plan changes are captured accurately.

SQL:

```
SELECT
  f.year_month           AS month,
  COUNT(DISTINCT f.org_key) AS pro_plus_orgs
FROM marts_marts.fact_subscription f
JOIN marts_marts.dim_plan p ON p.plan_key = f.plan_key
WHERE p.is_proplus = true
GROUP BY f.year_month
ORDER BY f.year_month;
```

Result:

Month	Pro+ orgs
2026-01	26
2026-02	29
2026-03	29
2026-04	29

Table 1: Monthly Pro+ subscriber count, January–April 2026.

What it means:

- The four values show the Pro+ trend since Rollouts launched on January 1, 2026.
- April’s count is also the denominator for the 60% Rollouts adoption target in Q2.
- The number is reproducible: one `dbt build` command gives the same result every time, unlike the prior manual CSV process.

Q2: Rollouts Adoption

The question: Of orgs currently on Pro+ (most recent month-end data, April 2026), the share that have fired at least one rollout-related notification at any point from January 2026 onward.

Grain: One row per (org, month-end) in `notif_count_month`, counting rollout-related notifications fired that month, with the denominator orgs coming from `fact_subscription` at the April 2026 month-end.

Model:

- `notif_count_month`: monthly fact, one row per org per month-end where the org fired ≥ 1 rollout-related notification. Measure: `rollout_notif_count`, which conforms to `dim_org` and `dim_date`.
- `fact_subscription` + `dim_plan`: supplies the Pro+ orgs at April 2026 month-end; this is the denominator.

How it works:

- A notification is rollout-related if its `event_type` starts with the prefix `rollout_`, `rollback_`, or `threshold_`. We use `LIKE 'rollout\_%' ESCAPE '\'` so the underscore is treated as a literal character, not a wildcard.
- All notification statuses are counted because the question asks whether a notification was fired, not whether it was successfully delivered.
- An org “adopted” Rollouts if it fired ≥ 1 such notification at any point from January through April 2026.
- The denominator is read from `fact_subscription`, not from current `billing_subscriptions`, so it reflects each org’s plan at April month-end.

SQL:

```
WITH
  april_pro_plus_orgs AS (
    SELECT DISTINCT f.org_key
    FROM marts_marts.fact_subscription f
    JOIN marts_marts.dim_plan p ON p.plan_key = f.plan_key
    WHERE f.date_key = 20260430
    AND p.is_proplus = TRUE
  ),
  orgs_with_rollout_activity AS (
    SELECT DISTINCT org_key
    FROM marts_marts.notif_count_month
    WHERE year_month BETWEEN '2026-01' AND '2026-04'
    AND rollout_notif_count > 0
  )
SELECT
  COUNT(DISTINCT a.org_key) AS pro_plus_orgs_april,
  COUNT(DISTINCT r.org_key) AS adopted_rollouts,
  ROUND(
    100.0 * COUNT(DISTINCT r.org_key) / NULLIF(COUNT(DISTINCT a.
      org_key), 0),
    1
  ) AS adoption_pct
FROM april_pro_plus_orgs a
LEFT JOIN orgs_with_rollout_activity r ON r.org_key = a.org_key;
```

Result:

Pro+ orgs (Apr 2026)	Adopted Rollouts	Adoption %
29	14	48.3%

Table 2: Rollouts adoption rate at April 2026 month-end.

What it means for Erin's brief:

Rollouts adoption among Pro+ orgs is 48.3% (14 of 29 orgs), below the 60% target Erin set.

Fewer than half of the organizations paying for Tracer's Pro+ tier are actively using its flagship feature, Rollouts. Since the Pro+ pricing premium depends on Rollouts adoption, this low engagement is a key risk for the CPO review. Specifically, 15 out of 29 Pro+ customers have not used Rollouts at all since its launch, which undermines both the renewal argument and the justification for the higher price point.

This aligns with the Q3 revenue trend: Pro MRR has remained flat since February with no movement between tiers. The organizations not adopting Rollouts are the same ones not contributing to Pro+ growth. The 48.3% adoption rate directly explains the stagnant

revenue. Since the upgrade thesis relies on feature usage, and over half of Pro+ customers are not using Rollouts, growth is stalled.

Q3: Revenue by Tier

SQL:

```
SELECT
  p.tier AS plan_tier,
  f.year_month,
  SUM(f.mrr_amount) AS total_mrr
FROM marts_marts.fact_subscription f
JOIN marts_marts.dim_plan p ON p.plan_key = f.plan_key
WHERE f.year_month >= '2026-01'
GROUP BY p.tier, f.year_month
ORDER BY f.year_month, p.tier;
```

Results:

Tier	Jan 2026	Feb 2026	Mar 2026	Apr 2026
Enterprise	\$130,725	\$134,709	\$134,709	\$134,709
Pro	\$48,269	\$50,955	\$50,955	\$50,955
Starter	\$9,193	\$7,743	\$8,062	\$7,308
Free	\$0	\$0	\$0	\$0
Total	\$188,187	\$193,407	\$193,726	\$192,972

Table 3: Monthly MRR by plan tier, January–April 2026.

What it means for Erin’s brief:

Total MRR has remained essentially flat since February, hovering around \$193,000 per month. Enterprise remains the dominant revenue tier, generating \$134,709 in April and accounting for approximately 70% of total MRR. While Enterprise revenue has remained stable, it has shown neither meaningful growth nor significant churn.

Pro MRR has also plateaued at \$50,955 since February. This metric is particularly important because the Rollouts feature, launched on January 1, 2026, was expected to drive upgrades from Starter to Pro. If Rollouts were successfully encouraging adoption and tier migration, Pro MRR would be steadily increasing. Instead, the only noticeable change was a modest \$2,686 increase between January and February, with no further growth afterward.

Meanwhile, Starter MRR has steadily declined from \$9,193 in January to \$7,308 in April, representing a 21% decrease over four months. The key question is whether these customers are upgrading to higher tiers or leaving the platform altogether. Revenue data alone cannot distinguish between these outcomes; analysis of the subscription change log is required to determine the cause.

As expected, the Free tier contributes no revenue.

For the CPO review, Pro+ MRR (Pro + Enterprise) totaled \$185,664 in April and has remained stable throughout the period. However, stability is not the growth narrative Tracer was expecting following the Rollouts launch. The feature was intended to shift customers into higher-value tiers, but the absence of Pro growth suggests that this transition is not occurring at scale. This finding aligns with the adoption analysis from Q2: organizations that have not adopted Rollouts are also the ones that have not upgraded, indicating that limited feature adoption is likely constraining revenue growth.

Recommendation

Recommendation: Run targeted Rollouts activation for the 15 non-adopting Pro+ orgs before their renewals.

Rollouts adoption sits at 48.3%, 12 points below the 60% target. Pro+ subscriber count grew from 26 to 29 between January and February, then flatlined. Pro MRR followed the same pattern, plateauing at \$50,955 since February with no further growth. Starter MRR declined 21% over the same period (from \$9,193 to \$7,308) without a corresponding rise in Pro, indicating those customers are leaving rather than upgrading.

The 15 Pro+ orgs with zero rollout activity are the highest-leverage target before the CPO review. An org that has never used the feature the tier is priced around has no reason to renew at the same rate. Tracer should identify these orgs from the Q2 adoption data, prioritize by contract age, and run direct activation outreach now. Closing the gap from 48% to 60% requires flipping roughly 4 orgs, which is achievable before year-end renewals if outreach starts immediately.

Extra Credit: Metabase Dashboard

A Metabase dashboard was designed and built against the marts to present all three numbers on a single screen. The mockup was designed first to define what each panel shows and how they compose together, then built live against `marts_marts`.



Figure 2: Dashboard mockup: panel layout for Rollouts adoption gauge, Pro+ subscriber trend, and MRR by tier.

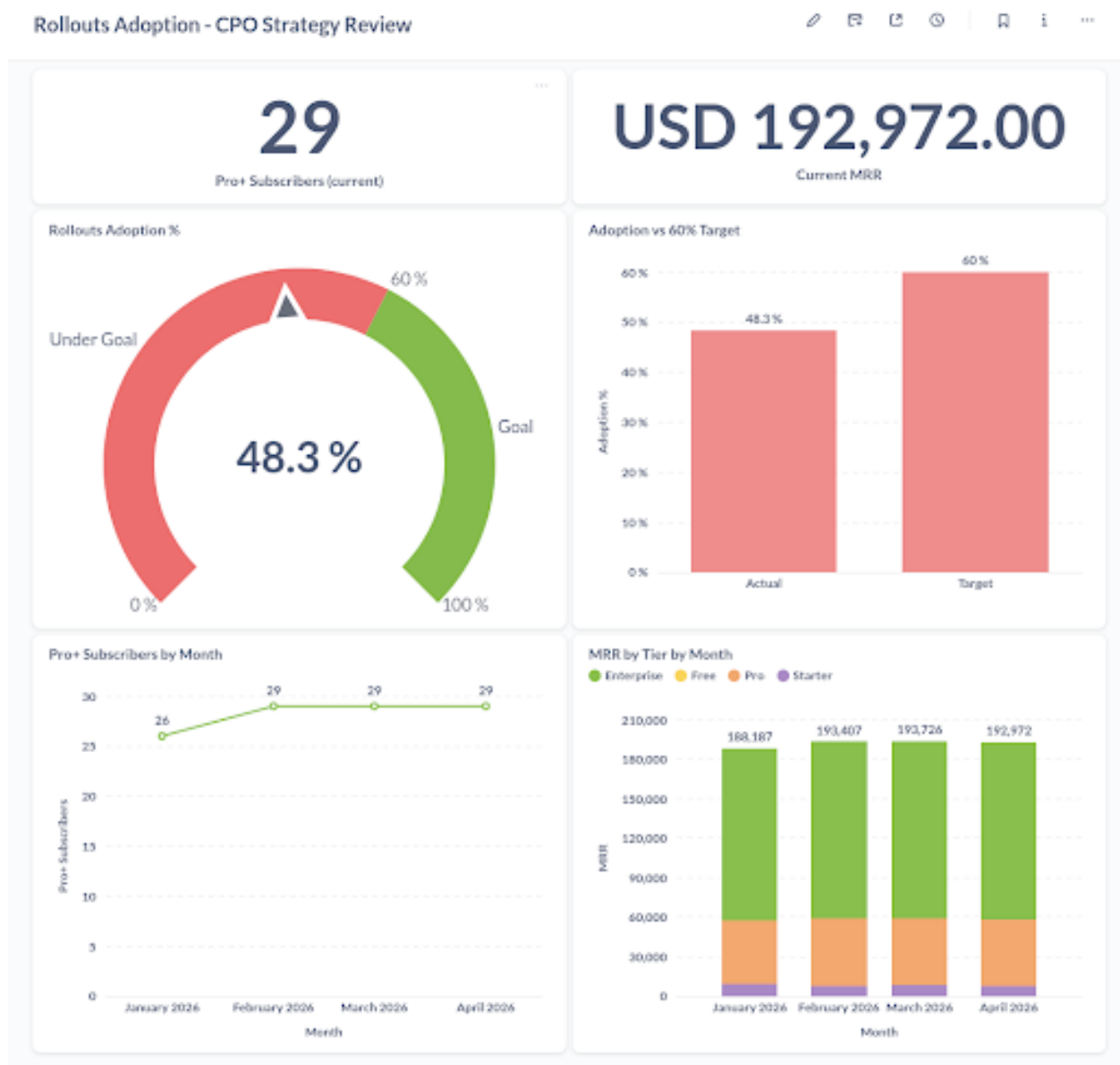


Figure 3: Dashboard actual: live Metabase dashboard built against `marts_marts`, showing 29 Pro+ subscribers, \$192,972 current MRR, and 48.3% Rollouts adoption vs. the 60% target.